Nevò **MIRZAI HAMADANI**

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PROFESSIONAL SUMMARY

Robotics master's student at EPFL with deep **simulation expertise**, from modifying physics engine internals (adding fluid curing and expansion in Genesis) to training RL and **imitation learning** policies across Isaac Sim, MuJoCo, and Gymnasium. Currently planning the development of imitation learning pipelines at Embodied AI for humanoid domestic tasks, after having built part of the teleoperation infrastructure. Research interests in sim-to-real transfer, imitation learning, world models and inverse reinforcement learning, for which I derived explicit error bounds on reward recovery under approximate dynamics.

EDUCATION

EPFL | *Master in Robotics (2nd year)* Sep. 2024 – December 2026
- Minor in Data Science: Reinf. Learn.: 6/6, Deep Learn.: 5.75/6, Image analysis: 5.75/6
- Relevant courses: Robotic Manip.: 5.5/6, Aerial Robotics: 5.5/6, Legged Robots: 5.25/6

EPFL | *Bachelor in Mechanical Engineering* Sep. 2021 – July 2024
- 1 year exchange at the Chinese University of Hong Kong (CUHK)
- Relevant courses: Linear Algebra: 5.75/6, Programming: 5.75/6, Algorithms: 5.75/6

EXPERIENCE

Embodied AI | *Robotics Internship, Humanoid Learning d* Feb. 2026 – Aug. 2026
- Developing **progress-aware policies** for long-horizon bimanual manipulation, combining **SARM** reward models with **DINO-WM** world model architecture to improve reliability on smartphone charging cable insertion task.
- Building teleoperation infrastructure for **imitation learning** datasets.

EPFL | *Teaching Assistant, Student Mentor* Sep. 2022 – Dec. 2025
- Teaching Assistant for Reinforcement Learning, Algorithms, Programming, Linear Algebra, and Mechatronics; Student Mentor for freshmen.

EPFL Rocket Team | *Project Lead: Flight Simulation* Sep. 2021 – July 2025
- Led flight simulation team developing vertical landing dynamics models in Simulink.
- Explored a control method approach utilising a MPC and neural networks for better stabilization and path planning for accurate landing. Trained and tested on Isaac Sim.

RESEARCH PROJECTS

Identifiability in Inverse Reinforcement Learning — RL Course, EPFL 6.0/6.0
- Surveyed three seminal works on **IRL identifiability** (Kim et al., Cao et al., Rolland et al.) covering graph-theoretic, entropy-regularized, and multi-expert settings.
- Derived an explicit **non-asymptotic sup-norm bound** on the reward estimation error as a function of the spectral deviation between true and empirical transition matrices, extending findings from Rolland et al.
- Provided **sample complexity bounds** for reward recovery.

3D Printing Drone Simulation — EMPA Lab Zurich, Prof. Kovac Paper Pending

- Built a simulation framework for an aerial additive manufacturing drone in the **Genesis** physics engine.
- Modified the engine's core **MPM (Material Point Method)** solver to model **viscoelastic curing and expanding fluid dynamics** — beyond standard rigid-body simulation.
- Validated simulated material deposition behavior against physical experiments.

Aerial Swarm Control via Hand Gestures — LIS Lab, EPFL, Prof. Floreano

- Designed an intuitive hand-gesture interface for controlling a drone swarm via a VR headset, with a modified **Olfati-Saber flocking algorithm** for swarm coordination.
- Built a simulated swarm in Unity and transferred control to a **real Crazyflie swarm via ROS2** (Sim-2-Real).

SELECTED PROJECTS

Imitation Learning with Robotic Arm — ACT, Vision-Based Control

- Assembled a leader-follower robotic arm system for teleoperated data collection.
- Implemented an **ACT (Action Chunking with Transformers)** policy for multi-step manipulation.
- Designed a vision-based control pipeline with **ResNet-18** encoders processing dual camera streams.
- Achieved ∼**85% success rate** on pick-and-place tasks from 50 demonstrations.

Reinforcement Learning for Quadruped Locomotion — PPO, Stable-Baselines3

- Compared **PPO** and **SAC**; trained the final **PPO** policy for quadruped locomotion.
- Achieved multi-gait control (trot, walk, pace, bound) with velocity tracking and **stair climbing** up to 5 cm step height.
- Designed **reward functions** for forward progress, smoothness, and drift penalty; used **adaptive terrain curriculum** for progressive difficulty.

Autonomous Drone Racing — Sim2Real, Path Planning

- Programmed autonomous trajectory racing in simulation, then deployed on a real Crazyflie via **sim-to-real transfer**.
- Integrated path planning, state estimation, and computer vision.

Data Creation Box — Object Recognition, System Integration - Winner, Startup Competition

- Built an automated physical environment generating diverse training datasets; fine-tuned object recognition models deployed on a robotic arm.

TECHNICAL SKILLS

Programming Python, C++, C, MATLAB **Technical** ROS2, Docker, CUDA, Git

Simulation IsaacSim, Mujoco, Gazebo, Simulink, Gymnasium, Unity, Genesis, Webots

Machine Learning SAC, PPO, ACT, Behavioral Cloning, IRL, PyTorch, IsaacLab

Computer Vision SLAM (Visual/LiDAR), Image Analysis, OpenCV

Laboratory Soldering, electrical circuits, CAD, 3D Printing

AWARDS - INTERESTS - LANGUAGES

World Record Academy (2017): Most drones piloted simultaneously by children

Talent Kick (2026): Selective cross-university high-tech startup incubator.

English C1, French C1, Italian C2, German B1 Judo (competed internationally)